3AL Tutorial 9 2024 Solutions

1. (a) Let A and B be distinct subgroups of a group G. Show that if $|A| = |B| = p$ then $|AB| = p^2$, that is, every element of the form ab (where $a \in A$ and $b \in B$) is distinct.

   HINT: Suppose there are some $a_1b_1, a_2b_2 \in AB$ such that $a_1b_1 = a_2b_2$. Rearrange this to find an element $x \in A \cap B$. Show that x must be the identity and so $a_1 = a_2$, $b_1 = b_2$.

Take elements $a_1b_1, a_2b_2 \in AB$ where $a_1, a_2 \in A$ and $b_1, b_2 \in B$. Suppose that $a_1b_1 = a_2b_2$. Then $a_2^{-1}a_1 = b_2b_1^{-1}$. Let this element be represented by x. Since $a_1, a_2 \in A$, we have $x \in A$. Similarly, $x \in B$ and so $x \in A \cap B$. Note that $A \cap B$ is a subgroup of A and so $|A \cap B| = 1$ or p. It is impossible for $|A \cap B| = p$ since A and B are distinct. Thus $A \cap B = \{1\}$. Thus $x = a_2^{-1}a_1 = b_2b_1^{-1} = 1$ meaning that $a_1 = a_2$ and $b_1 = b_2$. Therefore, each element of the form $a'b'$ is distinct and so $|AB| = p^2$.

   (b) Let G be a group of order pq for some primes p and q with $p > q$. Let $a \in G$ be an element of order p.

   (i) Why does such an element a exist?

   (ii) Use (a) to show that the only subgroup of G of order p is $A = \langle a \rangle$.

   (iii) Prove that $A \lhd G$.

(i) Because p is a prime that divides $|G|$, Cauchy's Theorem tells us there is an element of G of order p.

(ii) Suppose there is a subgroup B of order p that is distinct from A. Then $|AB| = p^2$ from (a). However, $AB \subseteq G$ and G has $pq < p^2$ elements. Therefore, we have a contradiction. This means that no such subgroup $B \neq A$ of order p can exist and so A is the unique subgroup of order p.

(iii) Let $g \in G$. Recall that G acts on the set of its subgroups by conjugation.

Therefore, gAg^{-1} is a subgroup of G. Moreover, gAg^{-1} has p elements. Since A is the unique subgroup of order p, we must have $gAg^{-1} = A$. Thus A is normal in G.

(c) Prove or disprove: There is a simple group of order 15.

NOTE: Recall that a group is called simple if its only normal subgroups are itself and its trivial subgroup.

Let G be a group of order $15 = 3 \times 5$. By Cauchy's Theorem, there is an element a of order 5. By (b), $\langle a \rangle \triangleleft G$. Thus G is not simple.

2. Let G be a p-group. Use the Class Equation to prove that $Z(G)$ is nontrivial.

Recall the Class Equation:

$$|G| = |Z(G)| + \sum_{i=1}^{n} |G : C_G(a_i)|.$$

Each index divides $|G|$, which is p^n, for some $n \geq 1$. Thus index must equal p^k for some k satisfying $1 \leq k \leq n$. Note that we cannot have $k = 0$ since that would mean $C_G(a_i) = G$ and then $a_i \in Z(G)$. Then p divides $|G|$ and each index $|G : C_G(a_i)| = |G|/|C_G(a_i)|$. Thus, p must also divide $|Z(G)|$. Therefore, $|Z(G)| > 1$ (it is impossible for $|Z(G)| = 0$) and so $Z(G)$ is nontrivial.

3. Let G be a non-abelian p-group. Show that G has an inner automorphism of order p.

HINT: The order of an inner automorphism is just the order of the inner automorphism when considered as an element of $\mathrm{Inn}(G)$. Recall that $\mathrm{Inn}(G) \cong G/Z(G)$.

Let G be a non-abelian nontrivial p-group. Then $|G| = p^n$ for some $n > 0$. Since G is a nontrivial p-group, $Z(G)$ is nontrivial. Moreover, since G is non-abelian,

$G \neq Z(G)$. Thus $|Z(G)| = p^k$ for some $0 < k < n$. By Lagrange's Theorem, $|G/Z(G)| = p^n/|Z(G)| = p^{n-k}$. Therefore, $|\mathrm{Inn}(G)| = p^{n-k}$ where $0 < n-k < n$. Then $p \mid |\mathrm{Inn}(G)|$ and so Cauchy's Theorem says that there is an element of $\mathrm{Inn}(G)$ of order p.

4. Let G be a p-group. Suppose G acts on a set X, show that p divides $|X| - |X_f|$.

By the Orbit Equation, $|X| = |X_f| + \sum_{i=1}^{n} |G : S(x_i)|$, where the elements x_i are the representatives of the distinct nonsingleton orbits of X. Each index $|G : S(x_i)|$ is a divisor of $|G|$. Thus each index must be some power of p. Each of these powers must exceed 0, since if $|G : S(x_i)| = p^0 = 1$ then $G * x_i$ is a singleton orbit, a contradiction. Thus p divides each $|G : S(x_i)|$. So p divides $|X| - |X_f|$. That is, $|X| \equiv |X_f| \bmod p$.

5. Let G be a p-group and H a nontrivial normal subgroup of G. Show that $H \cap Z(G)$ is nontrivial.

   Hint: Consider the group action of conjugation by G on H.

We have previously shown that conjugation by G is a group action on H. To show that $H \cap Z(G) \neq \{1\}$, it suffices to show $|H \cap Z(G)| \neq 1$. Since G is a p-group, we have $|H| \equiv |H_f| \bmod p$. Moreover, since $|H|$ is a nontrivial subgroup of the p-group G, we must have $p \mid |H|$. Thus $0 \equiv |H_f| \bmod p$. However,

$$\begin{aligned} H_f &= \{h \in H : ghg^{-1} = h \text{ for all } g \in G\} \\ &= \{h \in H : gh = hg \text{ for all } g \in G\} \\ &= H \cap Z(G). \end{aligned}$$

Therefore, p divides $|H \cap Z(G)|$. Since H and $Z(G)$ are both subgroups of G, their intersection contains the identity so it is impossible for $|H \cap Z(G)| = 0$. Thus $1 < p \leq |H \cap Z(G)|$ and $H \cap Z(G) \neq \{1\}$.

6. Let G be a non-abelian group of order p^3 for some prime p. Prove that $|Z(G)| = p$ and $G/Z(G) \cong \mathbb{Z}_p \times \mathbb{Z}_p$.

   Hint: Recall that if $G/Z(G)$ is cyclic, then G is abelian.

Since $Z(G)$ is a subgroup of G, we must have either $|Z(G)| = 1$, p, p^2 or p^3. Since G is a p-group, its centre is nontrivial, so $|Z(G)| \neq 1$. Moreover, since G is non-abelian, we know $|Z(G)| \neq p^3 = |G|$. If $|Z(G)| = p^2$, then $|G/Z(G)| = p^3/p^2 = p$ and so $G/Z(G) \cong \mathbb{Z}_p$. This means $G/Z(G)$ is cyclic, which means G is abelian, a contradiction. Therefore, $|Z(G)| = p$.

By Lagrange's Theorem, $|G/Z(G)| = p^3/p = p^2$. Then $G/Z(G)$ is isomorphic to either $\mathbb{Z}_p \times \mathbb{Z}_p$ or $\mathbb{Z}_{p^2}$. If $G/Z(G) \cong \mathbb{Z}_{p^2}$, then $G/Z(G)$ would be cyclic and so G would be abelian, a contradiction. Therefore, we must have $G/Z(G) \cong \mathbb{Z}_p \times \mathbb{Z}_p$.

7. Let G be a non-abelian p-group. Show that p^2 divides $|G : Z(G)|$.

Let $|G| = p^n$ for some positive integer n. Since G is a p-group, $|Z(G)| \neq 1$. Moreover, $Z(G)$ is a subgroup of G, so we can write $|Z(G)| = p^k$ for some positive integer $1 \leq k < n$. By Lagrange's Theorem, $|G : Z(G)| = |G|/|Z(G)| = p^{n-k}$.

To show that p^2 divides $|G : Z(G)|$, we need to show that $n - k \geq 2$. Note that $n - k \neq 0$ since that would mean $|G| = |Z(G)|$ and then G would be abelian. So, it remains to show that $n - k \neq 1$. Suppose $|G : Z(G)| = |G/Z(G)| = p$. Then $G/Z(G) \cong \mathbb{Z}_p$ and so the factor group is cyclic. This means that G is abelian, a contradiction. Therefore, we must have $n - k \geq 2$ and so $p^2 \mid |G : Z(G)|$.

8. (a) Let G be a p-group. Show that the number of subgroups of order p^k is congruent modulo p to the number of normal subgroups of order p^k.

   (b) Let G be a p-group. Show that the number of non-normal subgroups of G is a multiple of p.

HINT: Define some group actions and use the fact that $|X| \equiv |X_f| \bmod p$.

(a) Let X be the set of subgroups of G with order p^k. Consider the group action by G on X defined by $g * H = gHg^{-1}$ for $g \in G$. You can check yourself that gHg^{-1} is a subgroup of G with order p^k. We claim that X_f consists of all normal subgroups of G of order p^k. It suffices to show that $H \in X_f$ if and only if $H \lhd G$. Indeed, $H \in X_f$ if and only if $g * H = H$ for all $g \in G$. That is, $H \lhd G$. Then $|X| \equiv |X_f| \bmod p$, so the statement has been proved.

(b) Let X be the set of subgroups of G and let G act on X by conjugation. We claim that X_f consists of all normal subgroups of G. Then the number of non-normal subgroups of G will be $|X| - |X_f|$. Indeed, $H \in X_f$ if and only if $gHg^{-1} = H$ for all $g \in G$. That is, $H \lhd G$. Since $|X| \equiv |X_f| \bmod p$, we must have that p divides $|X| - |X_f|$.

9. Let G be a group with normal subgroup K. Prove that if G/K is a p-group, then G has a normal subgroup of index p.

Say $|G/K| = p^n$. Since G/K is a p-group, it has a normal subgroup of order p^k for any k with $1 \leq k \leq n$. Let $\mathcal{H}$ be a normal subgroup of G/K with order p^{n-1}. By the Correspondence Theorem, we can write $\mathcal{H} = H/K$ for some subgroup H of G that contains K. Since $\mathcal{H}$ is normal in G/K, we have that H is normal in G too. It remains to show that $|G : H| = |G/H| = p$. We have $|\mathcal{H}| = |H/K| = p^{n-1}$. We need to relate this to $|G/H|$. By the Third Isomorphism Theorem,

$$(G/K)/(H/K) \cong G/H.$$

Thus $|G/H| = |G/K|/|H/K| = p^n/p^{n-1} = p$. So H is a normal subgroup of G with index p.

10. What are the orders of the Sylow p-subgroups of a group of order 180? What about a group of order 700?

For a group of order 180, a Sylow 2-subgroup will have order 4, a Sylow 3-subgroup will have order 9, a Sylow 5-subgroup will have order 5.

For a group of order 700, a Sylow 2-subgroup will have order 4, a Sylow 5-subgroup will have order 25, a Sylow 7-subgroup will have order 7.

11. Find all Sylow p-subgroups of S_3 for $p = 2$ and $p = 3$. Confirm the following (for $p = 2$ and $p = 3$, separately):

    (a) Any two Sylow p-subgroups are conjugate.

    (b) The number of Sylow p-subgroups is congruent to 1 modulo p.

    (c) The number of Sylow p-subgroups divides $|S_3|$.

12. Let G be a group with order $p^n m$ where p is a prime with $p \nmid m$. Suppose H is a Sylow p-subgroup of G. Prove or disprove: Any conjugate of H is a Sylow p-subgroup.

Since H is a Sylow p-subgroup, we have $|H| = p^n$. Let $g \in G$. Consider the inner automorphism $\sigma_g : G \to G$. Then $\sigma_g(H) = gHg^{-1}$. Since σ_g is an isomorphism, $|H| = |\sigma_g(H)| = |gHg^{-1}|$. Thus $|gHg^{-1}| = p^n$. Moreover, since σ_g is an automorphism, gHg^{-1} is indeed a subgroup of G. Thus, it is a Sylow p-subgroup as desired.

NOTE: This proof can be done more directly (without involving the inner automorphism).